

Python Programming

Day 1 – Introduction & Basics

Duration: 2 Hours

Level: Beginner

Goal: Understand what Python is and write your first Python programs.

1 What is Python? (15 minutes)

Python is a high-level, interpreted programming language that is:

- Easy to read and write
- Used in web development, data science, AI, automation, games, and more

Why Python?

- Beginner-friendly syntax
- Large community
- Thousands of libraries

Common Uses

- Web apps (Django, Flask)
- Data Science (Pandas, NumPy)
- AI & ML
- Automation & scripting

2 Setting Up Python (10 minutes)

Check if Python is installed

```
python --version
```

or

```
python3 --version
```

Tools

- Python Interpreter
- VS Code / PyCharm
- Jupyter Notebook (optional)

👉 For class, you can run Python using:

- Terminal
- VS Code
- Online compiler (Replit, Programiz)

3 Your First Python Program (15 minutes)

Example: Hello World

```
print("Hello, World!")
```

Explanation

- `print()` displays output
- Text is written inside quotes

4 Comments in Python (10 minutes)

Comments are notes ignored by Python.

```
# This is a single-line comment
```

```
"""  
This is a  
multi-line comment  
"""
```

Why use comments?

- Explain code
- Make programs easier to understand

5 Variables & Data Types (30 minutes)

Variables

A variable stores data.

```
name = "Alice"  
age = 25  
height = 5.6  
is_student = True
```

Rules

- No spaces
- Cannot start with numbers
- Use meaningful names

Data Types

Type	Example
int	10 , -5
float	3.14 , 2.0
str	"Hello"
bool	True , False

Check type:

```
type(age)
```

6 User Input & Type Conversion (20 minutes)

Taking Input

```
name = input("Enter your name: ")  
print("Hello", name)
```

⚠ Input is always a **string**

Convert Input

```
age = int(input("Enter your age: "))  
print(age + 5)
```

Other conversions:

```
float("3.5")  
str(100)
```

7 Basic Operators (20 minutes)

Arithmetic Operators

Operator	Meaning
+	Addition
-	Subtraction
*	Multiplication
/	Division
%	Modulus
**	Power

Example:

```
a = 10
b = 3
print(a + b)
print(a ** b)
```

8 Mini Practice (15 minutes)

Exercise 1

Write a program that:

- Takes user name
- Takes age
- Prints a message saying how old they will be in 10 years

```
name = input("Enter name: ")
age = int(input("Enter age: "))
print(name, "will be", age + 10, "years old in 10 years.")
```

Exercise 2

Create a simple calculator:

- Take two numbers
- Print sum, difference, product

Common Beginner Mistakes (10 minutes)

 Forgetting quotes:

```
print(Hello)
```

 Mixing strings & numbers:

```
print("Age is " + 20)
```

✓ Correct:

```
print("Age is", 20)
```

Day 1 Summary

Today you learned:

- ✓ What Python is
- ✓ How to run Python
- ✓ Variables & data types
- ✓ Input & output
- ✓ Basic operators



Homework (Optional)

1. Write a program that converts Celsius to Fahrenheit
2. Write a program that calculates the area of a rectangle